



**IN
PARTNERSHIP
WITH
PLYMOUTH
UNIVERSITY**

Software Development Tools and Practices PUSL2020

Coursework 2025–2026

Semester: Semester 2

Submission Deadline: TBA

Coursework Type: Group Assignment

Element of Assessment: C1

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1 Coursework

This coursework for PUSL2020 is worth 50% of the module marks. It is made up of two parts (i) a software product and (ii) a report containing your testing strategies and implementation. Students must create their own groups, up to a maximum of six group members. Each group must then decide and agree upon which technology they will use as well as specifying and allocating individual roles, and responsibilities.

Note: Each member should at least implement one functionality.

2 Project Goal

The goal of this coursework is to design and develop a web-based dashboard that provides real-time air quality information for the Colombo Metropolitan Area. The dashboard will visualize data from simulated air quality sensors, allowing users to monitor Air Quality Index (AQI) levels across Colombo, explore historical trends, and understand the spatial distribution of air quality.

3 Project Scenario

Air pollution is a growing concern in Colombo, impacting public health and the environment. To address this, the Colombo Municipal Council aims to implement a real-time air quality monitoring system and make this information publicly accessible.

This coursework involves developing a **Real-time Air Quality Monitoring Dashboard for Colombo**. The platform will simulate data from a network of air quality sensors strategically placed across Colombo. The dashboard will display this simulated data on a map, providing real-time AQI readings and historical trends (as seen in figure 01). This system will empower environmental agencies, researchers, and the public to stay informed about air quality in Colombo and support data-driven actions for a cleaner environment.

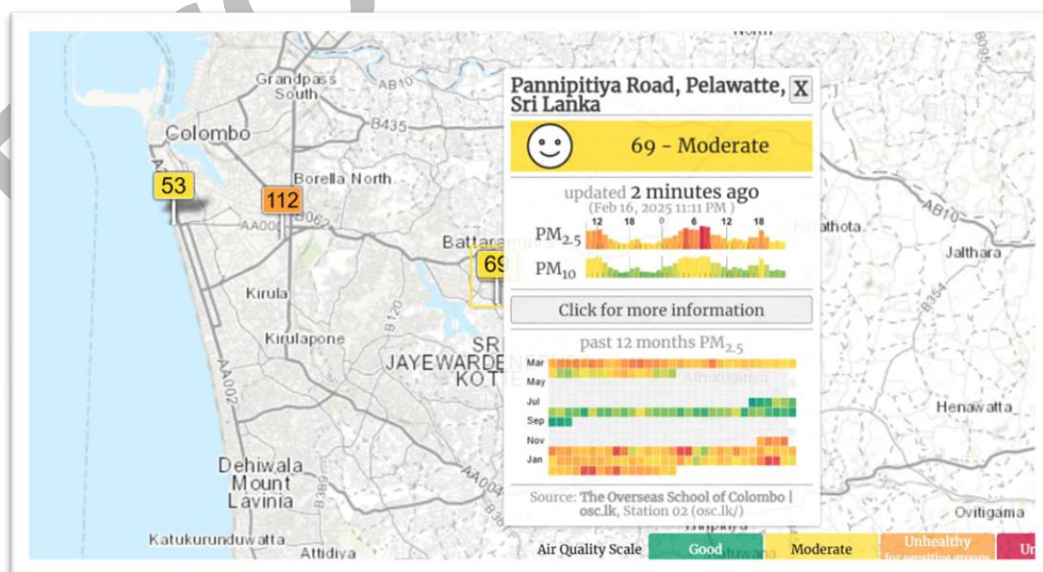


Figure 1. Sample GUI where the AQI readings and historical trends in a map.

4 User Roles and Functional Requirements

4.1 Monitoring Admin (Environmental Agency/ System Admin)

Responsibilities: System configuration, sensor network management (simulated), data feed management (simulated), user account management, setting alert thresholds, validating simulated data quality, system maintenance.

Functional Requirements:

- Secure Login.
- Sensor Management (Simulated):
- Register new simulated sensors (add location within Colombo on a map, sensor ID, and other fields if necessary).
- Manage simulated sensor details (view, edit, deactivate).
- View simulated sensor status (active/inactive).

Data Simulation Management:

- Configure data simulation parameters (e.g., frequency of data generation, baseline AQI levels, potential variation patterns - basic level).
- Start/Stop data simulation.

Alert Configuration:

- Set alert thresholds for AQI levels (e.g., levels for "Moderate," "Unhealthy" air quality categories).
- Manage alert notification display on the dashboard.

User Management:

- Create and manage accounts for Monitoring Admins.

System Dashboard:

- View system status, simulation status, number of active sensors.

4.2 Data Provider (Simulated Sensor Data Generator)

Responsibilities: Automatically generate realistic, but simulated, air quality data for each sensor location in Colombo.

Functional Requirements (Background Simulation Process):

- Automated Data Generation: The system must include a background process that automatically generates simulated AQI data for each registered sensor at regular intervals (e.g., every 5-15 minutes).

Data Persistence: Simulated data should be stored in the database for historical data access.

NOTE: You can use real-time data sources(<https://waqi.info/>, <https://map.purpleair.com/>) or synthetic data generation for this. The attached data.csv file provide you with a basic structure for synthetic data generation.

4.3 Public User/Citizen

Responsibilities: View real-time and historical air quality information for Colombo.

Functional Requirements:

- Public Access (no login required).

Map View:

- Display simulated sensor locations on a map of Colombo.
- Use color-coded markers to represent current AQI levels at each sensor location (e.g., Green for Good, Yellow for Moderate, Orange/Red for Unhealthy – use standard AQI color scale).

Click on a sensor marker to:

- Display current AQI reading.
- Display a small pop-up chart showing recent historical AQI trend (e.g., last 24 hours).

Historical Data Visualization:

- Allow users to select a sensor location from the map or a list.
- Display interactive charts showing historical AQI data for the selected sensor over different time periods (e.g., last day, last week, last month - focus on daily trends initially).

Legend:

- Display a clear legend explaining the AQI color codes and categories (e.g., "Good," "Moderate," "Unhealthy").

4.4 System Administrator (Web Master)

Responsibilities: System infrastructure management, server setup (local development/potential deployment environment), database management, basic security configurations, and overall system health for the coursework setup.

Functional Requirements:

- User account management (for Monitoring Admins).
- System configuration (database settings, etc.).
- Basic security setup for a coursework project.

5 Deliverables

There are two deliverables, and both should be submitted via DLE.

5.1 Deliverable 1 [Development]

Develop a web application for a Real-time Air Quality Monitoring Dashboard for Colombo. Use PHP (Laravel recommended) or ASP.NET C# (ASP.NET Core recommended) **ONLY (other frameworks will not be allowed)** for backend, and MySQL, PostgreSQL, or SQL Server (or similar) for the database. Utilize standard web frontend technologies (HTML, CSS, JavaScript) and a mapping library (e.g., Leaflet, Google Maps API).

5.2 Deliverable 2 [Test and Report]

Write a comprehensive report comprising 2000 words explaining your testing strategies. The report should include:

- Evidence of your development (screenshots of all the pages along with a summary and the list of developers at the bottom of each screenshot)
- Breakdown of the individual contribution of the team members. (with the implemented functionality)
- A description of the test cases you have designed
- A description of the structure and role of any mock objects you have used
- An explanation of how to run the unit tests and the integration tests
 - Your functional test plans
 - A critical analysis of your test strategy, explaining why you have tested the software in the way that you have.

6 Detailed Assessment Criteria

Category and Weighting	Criteria	Marks	LOs
Software Development 50%	• Application should satisfy all functional requirements.	25	LO1
	• Code Quality and Best Practices	15	
	• Database Integration and Data Handling	10	
Testing and Reporting 50%	• Report Clarity and Professionalism	15	LO2
	• Comprehensive Testing Implementation	20	LO3
	• Test Strategy Justification and Analysis	15	LO4

6.1 Marking Rubric:

Mark Range	Software Development (25%) - Functionality & Requirements Satisfaction (Max 50)	Software Development (15%) - Code Quality & Best Practices (Max 30)	Software Development (10%) - Database Integration & Data Handling (Max 20)	Testing and Reporting (15%) - Report Clarity & Professionalism (Max 30)	Testing and Reporting (20%) - Comprehensive Testing Implementation (Max 40)	Testing and Reporting (15%) - Test Strategy Justification & Analysis (Max 30)
Below 40 (Fail)	Very few requirements met. Core features missing/non-functional.	Poor code, unstructured, minimal best practices understanding.	Minimal/non-functional DB integration, flawed data handling.	Unclear, unprofessional report, poor structure, minimal evidence.	Minimal/absent testing, no unit/integration/functional tests.	No test strategy/analysis, inadequate justification.
40-50 (Pass - Basic)	Some core requirements met, significant flaws/incompleteness.	Functional code, poorly structured, lacks validation/security, inconsistent practices.	Basic DB integration, flawed insert/edit/delete etc.	Understandable report, lacks clarity/professionalism, basic structure, incomplete evidence.	Basic testing attempts, limited scope/coverage, unclear results.	Basic test strategy attempt, weak justification/analysis.
50-60 (Pass - Satisfactory)	Most core requirements met, minor flaws/incompleteness.	Reasonably functional/structured code, some validation/security, adequate practices.	Satisfactory DB integration, some robustness issues in data handling.	Generally clear/professional report, adequate evidence with summaries.	Satisfactory unit/integration/functional tests, reasonable clarity.	Reasonable strategy, basic analysis, demonstrates basic understanding.
60-70 (Good -)	All core requirements	Good code quality, well-	Well-implemented/r	Clear, professional,	Good unit/integration/fu	Well-defined strategy,

Above Average)	met well, mostly well-implemented.	structured, good validation/security, consistent practices.	robust DB integration, efficient/reliable data handling.	well-structured report, logical argument, well-presented evidence.	functional tests, good coverage, clear explanations.	effective analysis, good testing understanding.
70+ (Excellent - Outstanding)	All requirements fully met/exceeded, robust, feature-rich, excellent functionality.	Excellent code quality, highly structured, exemplary validation/security, best practices.	Exceptional DB integration, highly efficient/robust data handling, advanced techniques.	Exceptionally clear/professional report, insightful argument, comprehensive & professional evidence.	Comprehensive/rigorous testing, exceptional coverage/thoroughness, meticulous documentation.	Insightful/well-justified strategy, sophisticated analysis, strong testing understanding.

7 Academic Offences

See below for excerpt of Section AST10.2. Full details on the academic offences framework and procedures, consult Section AST10 is [available here](#).

Academic offences occur when activity is undertaken which could confer an unfair advantage to any candidate(s) in assessment. The University recognises the following (including any attempt to carry out the actions described) as academic offences, regardless of intent:

- Copying or paraphrasing of other people's work or ideas into a submitted assessment without full acknowledgement (plagiarism).
- Unauthorised collaboration of students (or others) in a piece of work (collusion).
- Making false declarations in an attempt to obtain either modified assessment provisions or special consideration (e.g. of extenuating circumstances).
- Persuading another member of the University or partner institution (student, staff, or other) to participate in any way in actions which would be in breach of these regulations.
- Misrepresenting research outcomes and results.
- Being party to any arrangement which would constitute a breach of these regulations.
- The inclusion in a piece of assessed work (other than an examination or test) of material which is identical or substantially similar to material which has already been submitted for any other assessment within the University.
- Any other activity which could confer an unfair advantage to any candidate(s).

8 Submission Type

The report should be a PDF (preferred) or Word document. A zip file containing project work and other assets should be uploaded to the Plymouth Digital Learning Environment (DLE).